

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1504

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 25

Question Count: 5

Code of Conduct

I **Edrin J S** (Name / Reg No) **192524041** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Edrin J S*

Date: 29 July 2026 **Batch:** 2025

Criteria	Relevance to Application (1.25)	Use of Cloud Concepts (1.25)	Practical Insight (1)	Completeness (0.75)	Clarity (0.75)	TOTAL (5)
Weightage	25%	25%	20%	15%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

APPLICATION BASED QUESTIONS — ANSWERS

1. How does cloud federation improve resource sharing and availability across datacenters?

Cloud federation connects multiple independent cloud providers or datacenters into a single, interoperable system, allowing resources such as compute, storage, and network capacity to be shared across organizational boundaries. This improves resource sharing because one provider can temporarily borrow capacity from another during peak demand (cloud bursting), balancing load more efficiently than any single datacenter could alone. It improves availability through geographic redundancy — if one datacenter fails or becomes overloaded, workloads can be redirected to a federated partner site — supporting disaster recovery, reduced latency for regional users, and greater overall resilience of cloud services.

2. What are the key security issues in virtualization, and how can they be mitigated?

- Hypervisor vulnerabilities and VM escape: A compromised hypervisor could let an attacker break out of one VM and access others or the host. Mitigation: keep hypervisors patched, minimize their attack surface, and use hardware-assisted virtualization security.
- Data leakage and side-channel attacks between co-located VMs: Shared physical resources can be exploited to infer data from neighboring tenants. Mitigation: strong resource isolation, encryption of data at rest and in transit, and dedicated hosts for highly sensitive workloads.
- Insecure or outdated VM images and VM sprawl: Unmanaged or unpatched VM images increase the attack surface. Mitigation: use scanned, hardened golden images, enforce VM lifecycle management, and regularly patch templates.
- Weak access control over the virtualization management layer: Compromised hypervisor credentials can affect the entire environment. Mitigation: role-based access control, multi-factor authentication, and detailed audit logging of administrative actions.

3. How does a software-defined datacenter enhance scalability and management?

A Software-Defined Datacenter (SDDC) abstracts compute, storage, and networking resources into software-controlled pools (via server virtualization, software-defined networking, and software-defined storage) that are decoupled from the underlying physical hardware and managed through a centralized control plane. This enhances scalability because resources can be provisioned, resized, or decommissioned on demand through automation and APIs, without manual hardware reconfiguration, allowing the infrastructure to grow or shrink elastically with workload needs. It enhances management because administrators can apply consistent, policy-based configuration, security, and monitoring across the entire environment from a single interface, reducing manual errors, enabling faster deployment, and allowing dynamic workload placement and load balancing across the datacenter.

4. How can identity and access control ensure data privacy in cloud-based healthcare systems?

In cloud-based healthcare systems, identity and access management (IAM) protects patient data privacy through several layered controls: role-based access control (RBAC) ensures only authorized clinicians or staff can view records relevant to their role; multi-factor authentication strengthens login security beyond passwords; and attribute-based access control can further restrict access based on context, such as department or assigned patients. Every access to protected health information (PHI) should be logged and audited to support compliance with regulations such as HIPAA or GDPR, while encryption of data at rest and in transit ensures that even if data is intercepted or storage is compromised, it remains unreadable. Identity federation also allows secure single sign-on across hospital systems while enabling administrators to quickly revoke access when a staff member's role changes or their employment ends, minimizing the window of unauthorized exposure.

5. How do virtualization and hypervisors support scalability and resource efficiency in cloud applications?

Virtualization abstracts physical hardware into multiple independent virtual machines, allowing several workloads to run on a single physical server instead of dedicating an entire machine to one application — this consolidation significantly increases resource utilization and efficiency while reducing the number of physical servers, cost, and energy consumption needed. The hypervisor manages the dynamic allocation of CPU, memory, and storage among VMs, enabling elastic scalability: VMs can be spun up, resized, or shut down quickly in response to changing demand, and features like live migration allow workloads to be moved between physical hosts for load balancing without downtime. Together, these capabilities let cloud applications scale up or down efficiently while making the most effective use of available hardware resources.